# Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

| Role | Name | Email |
|---|---|---|
| **Developer (You)** | Jake Pombo | Jpombo@udel.edu |
| **Friend (optional)** | James Hicks<br>Matthew Fitzgerald | JPhicks@udel.edu<br>Mafitz@udel.edu |

# Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea

and a paragraph describing how that paper supports your design concept.
        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider the project below.
        - Create an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the chempy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.

| |
|---|
| I am choosing the chemistry example from above. I like chemistry and believe that having a website that can find percent yield, limiting reactants and more from chemical reactions. |

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

| |
|---|
| This website: Limiting Reactant Calculator, calculates the limiting reactant by collecting the mass, coefficient, compound/element and unit and same for products and finds the limiting reactant. This website:Reaction Yield Calculator does the same thing but calculates the reaction yield. |

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

| |
|---|
| This website would be very useful for people in chemistry classes who want to learn or just need a quick calculator that can do math they may take some time. I am currently in a chemistry class and know how to calculate all these things but having to calculate all of |

them out can be tiring when trying to study or just do homework. This solves this.

# Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

This program will take in the arguments of compound/element, mass and coefficient, then return the limiting reactant, theoretical yield and percent yield.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

They will need to provide the compound/element, mass and coefficient of each reactant and product.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will be math with equations from general chemistry.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back the limiting reactant, theoretical yield and percent yield.

# Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

| |
|---|
| Mass, Compound/element, coefficient, product, reactant, percent yield, limiting reactant, theoretical yield. |

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

| |
|---|
| Mass number, compound/element name, coefficient number, product and reactant names, percent yield number, theoretical yield number and limiting reactant name. |

# Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

| |
|---|
| Having the code be able to take in a number of arguments in different functions and produce a number of things based on those arguments. |

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

| |
|---|
| I am going to be using an LLM. To make sure I avoid shallow learning, I will go over each bit of code and make sure I know how to do it if I were to write the whole program out myself. |

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

| |
|---|
| My ultimate goal is to create a program that can help me in the future with chemistry. I have |

also struggled in the past with creating full functioning websites so I would like to understand why they didn't work in the past.

# Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a website that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield. What problems do young college students normally face when solving these problems?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

AI said that a few common problems students may have are understanding balanced equations, converting from moles to grams or vise versa and identifying the limiting reactant.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

What is a less complex way to code in the molar masses and chemical information. Could you have the students type it in as a parameter?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

*Instead of having a dictionary of chemical weights and elements, they will now be parameters students will need to enter in themselves.*

# Part 6: Resource Planning

1. What resources will you use if you get stuck?

*Python textbook, TA's, Professors, AI*

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1)ask AI to check its code again
2)go through on thawny and use debug software
3)Then go line by line till you find the mistake and then fix it

# Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I would build this website in separate parts so you can understand and test each piece instead of trying to write everything at once. Since you're building a chemistry calculator, the most important part is getting the calculation logic correct before worrying about the design.

**1. Break the website into five components**

1. User interface (frontend)

Collect the balanced equation, starting mass of each reactant, desired product, and actual harvested mass.

Possible technologies: HTML, CSS, JavaScript, or React.

2. Chemical equation parser

Read the equation and identify the reactants, products, chemical formulas, and coefficients. It must distinguish coefficients from subscripts.

For example, $2H_2 + O_2 \rightarrow 2H_2O$ contains two reactants and one product.

3. Chemistry data and molar masses

Determine each substance's molar mass from its chemical formula and atomic masses. You could calculate these from a periodic-table dataset rather than manually entering every possible compound.

4. Calculation engine

Perform the gram-to-mole conversions, limiting-reactant comparison, theoretical-yield calculation, and percent-yield calculation.

5. Results and explanations

Show the final answers alongside the intermediate steps, units, and explanations of why a particular reactant is limiting.

**2. Build the calculation logic in the right order**

I'd implement the chemistry calculations as separate functions. That way, you can test each one independently.

| Function | Responsibility |
| --- | --- |
| Equation parser | Identify formulas and coefficients |
| Equation validator | Check that each element has the same atom count on both sides |
| Molar mass calculator | Calculate the mass per mole of each substance |
| Mass-to-moles converter | Convert each starting mass to moles |
| Limiting reactant calculator | Compare available amounts relative to coefficients |
| Theoretical yield calculator | Calculate the maximum possible product mass |
| Percent yield calculator | Compare actual and theoretical yields |

One important technical decision: don't calculate the limiting reactant by comparing grams. Convert to moles first, then account for the balanced equation's coefficients.

**3. Test the chemistry before building the interface**

Use a known reaction as a test case:

$$2H_2 + O_2 \rightarrow 2H_2O$$

Suppose you start with 4 g of hydrogen and 32 g of oxygen, and harvest 30 g of water.

Your program should be able to:

1. Convert the hydrogen and oxygen masses into moles.
2. Determine which reactant runs out first based on the 2:1 mole ratio.
3. Calculate the maximum mass of water that can form.
4. Calculate the percentage yield from the 30 g harvested amount.
5. Explain every conversion and calculation.

For testing, you should also try cases with excess reactants, invalid formulas, an unbalanced equation, and a harvested amount greater than the theoretical yield.

**4. Choose a development approach**

For a first version, I'd recommend HTML, CSS, and JavaScript if you want a simple website that runs directly in a browser. React is a good option if you want a more interactive interface with dynamic input fields and expandable explanations.

I'd avoid starting with a database or user accounts. You can calculate everything locally in the browser at first.

My biggest recommendation: build and test the calculation engine first, then connect it to the interface. The hardest part is reliably interpreting arbitrary chemical formulas and equations, not displaying the results.

simple python

I would start with a simple Python program using separate functions for each chemistry calculation. You don't need a website interface yet; first, get the calculations working in the terminal.

**1. Break it into steps**

1. Ask the user for the balanced equation.
2. Ask for the starting mass of each reactant.
3. Convert grams into moles using molar mass.
4. Determine the limiting reactant using the balanced coefficients.
5. Calculate theoretical yield.
6. Ask for the actual harvested product mass and calculate percent yield.

**2. Python concepts you'll need**

| Concept | How you'll use it |
| --- | --- |
| `input()` | Collect the equation and masses |
| Variables | Store coefficients, masses, and results |
| Dictionaries | Store atomic masses and chemical information |
| Functions | Separate the calculations into manageable parts |
| `for` loops | Process each reactant and element |
| `if` statements | Compare reactants and validate inputs |
| Arithmetic | Perform stoichiometric calculations |

**3. Build it in stages**

I'd start with just one reaction, such as:

$$2H_2 + O_2 \rightarrow 2H_2O$$

Get the program working for that reaction before trying to support every possible chemical equation. Otherwise, parsing chemical formulas and calculating molar masses will make your first project unnecessarily complicated.

Your first task: write a Python function that takes a mass in grams and a molar mass as inputs, then returns the number of moles.

Once that works, you can build the limiting-reactant calculation on top of it.

what is less complex way tom code in the molar masses and chemical information. could you have the students type it in as a parameter?

Yes! Having students enter the molar masses as parameters is a much simpler way to build your first version

in Python. You wouldn't need to create a database of every chemical compound or write code to interpret chemical formulas.

There are two good approaches.

**Option 1: Students enter molar masses manually**

For example, for the reaction

$$2H_2 + O_2 \rightarrow 2H_2O$$

The program asks the student to enter:

- Mass of $H_2$
- Molar mass of $H_2$ (approximately 2.016 g/mol)
- Mass of $O_2$
- Molar mass of $O_2$ (approximately 32.00 g/mol)
- Molar mass of $H_2O$ (approximately 18.015 g/mol)
- Actual harvested mass of water

Your program then performs the calculations using those values.

Advantages:

- Very beginner-friendly Python.
- No chemical formula parser needed.
- Works with many different reactions.
- Easy to debug because the inputs are explicit.

Disadvantage: Students can enter an incorrect molar mass, so you should validate that the values are positive numbers and explain where to find molar masses on the periodic table.

**Option 2: Students enter chemical information in a dictionary**

You could store common compounds in a Python dictionary, such as hydrogen, oxygen, and water, and retrieve their molar masses automatically. This reduces typing but requires you to maintain the list of supported substances.

For a beginner project, I'd choose Option 1.

**How I'd structure the Python program**

Without writing the code for you, I'd organize it into these functions:

| Function | Parameters | Purpose |
|---|---|---|

| | | |
|---|---|---|
| `get_reactant_data()` | Mass, molar mass, coefficient | Collect information for a reactant |
| `calculate_moles()` | Mass, molar mass | Convert grams to moles |
| `find_limiting_reactant()` | Reactant data | Compare mole amounts against coefficients |
| `calculate_theoretical_yield()` | Limiting reactant data, product molar mass | Calculate maximum product mass |
| `calculate_percent_yield()` | Actual yield, theoretical yield | Calculate percentage yield |

One important detail: students would still need to enter the balanced coefficients for each substance. The coefficients are essential for identifying the limiting reactant correctly.

I'd start by supporting reactions with two reactants and one product. Once that works, you can expand the program to support more reactants and products.

# Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.